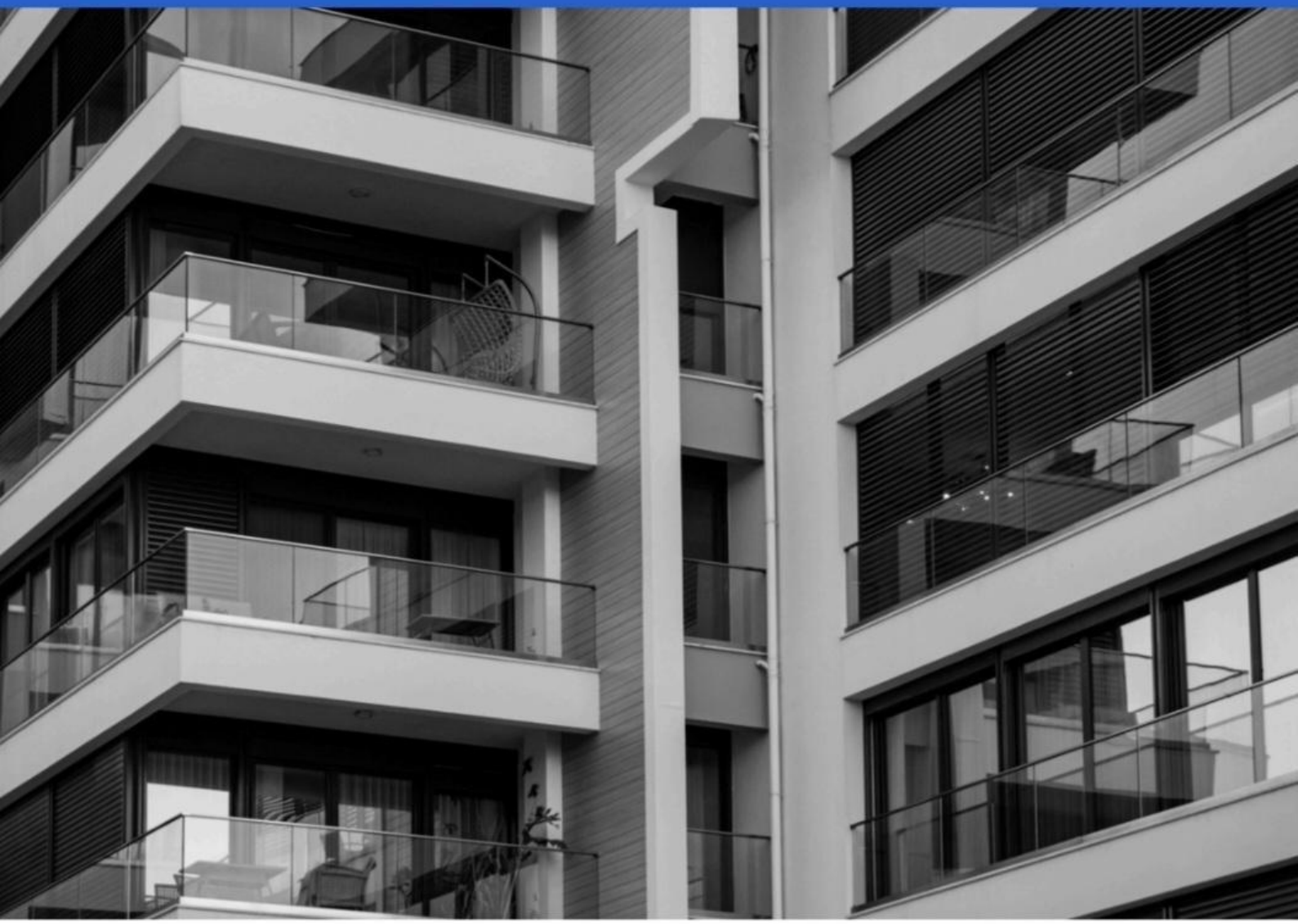
PRESENTED BY:
GROUP 8

SUPERVISOR:
MR. THABISO LUNGA APHANE



ANALYSIS PHASE

VALORA

PROPERTY INTELLIGENCE AND PRIVATE COMMUNITY MANAGEMENT HUB

Contents

Introduction..... 3
Information Gathering Methodology (Observation, Participatory, Interviews)..... 3
Analysis of Existing System..... 7
Data Analysis (Data Integrity C Constraints) 8
Weakness of Current System 12
Analysis of the Proposed System (Functional Requirements) 14
Non-Functional Requirements 15
Data Modelling for Proposed System..... 16

Introduction

VALORA is a new Private Property and Community Management platform tailored for housing complexes and gated communities in South Africa. It addresses huge gaps in the property market by integrating AI-driven property intelligence (valuations, growth scores, rental yields, and 10-year ROI forecasts) with an integrated community platform. It also enhances the decision making of real estate professionals, unlike pre-existing services. In addition to property data, VALORA also offers homeowners and residents centralized tools to report complaints, access trusted local service providers and work collaboratively with Homeowners Associations (HOAs). This digital consolidation allows for transparency, community involvement, and services to be delivered in the everyday life of the community. We have chosen to analyze four existing platforms that we believe are relevant for the design of VALORA. These platforms include Property24 and Private Property (public listing portals), Trafalgar (professional HOA management), and EstateMate (a community management mobile app).

The platform has different user roles such as residents having access to community features and property information; investors receiving financial risk information; estate agents handling listings and living scores; casual visitors being allowed to browse through properties and get basic valuations; and HOA admins being able to verify providers and moderate complaints. This role-based design ensures users are getting relevant, actionable information that will strengthen their interactions with their communities.

Developed with Agile Scrum over 14 bi-weekly sprints, VALORA's iterative approach takes user feedback into account and adapts it to AI features as they change. Community stakeholder engagement also helps to make the product more user-friendly and acceptable. VALORA democratizes property data and elevates community living with a simple, scalable, user-driven platform that addresses the real problems of the South African property market to support investment and the lives of residents.

Information Gathering Methodology (Observation, Participatory, Interviews)

Observation:

Observation is the process of watching and noticing how things work without interfering. This means a review of the current practice of how brokers, HOAs and residents list properties, find service providers, and deal with complaints. This will allow you to get a feel for what your platform is replacing, and possible irritations and solutions that interviewees may not have considered.

We have seen complaints being dealt with in the following ways: Residents email the HOAs, contact the management office or place them in a suggestion box. HOAs frequently have

their inboxes overflowing and are overwhelmed by them. The manager does not ignore the complaints out of spite but rather they struggle to do a proper job of prioritizing concerns. This causes the residents to become agitated due to prolonged waiting times for a response. In an attempt to catch the HOA's attention, this may lead them to file the same complaint several times. Residents don't have an immediate sense of urgency, which either results in them not communicating the actual level of urgency or underestimating the urgency, potentially increasing the overall property damage.

When asked about the kind of service providers that homeowners need, we found that the homeowners reach out to the members of the HOA, who have a hard copy or digital list of authorized service providers (plumber, electrician, landscapers, pool cleaners). Otherwise, they search online on Google or ask their family, friends and neighbours for suggestions. We saw that several residents have the following issues where sometimes the line is disconnected, and the HOA has its line of choice that is no longer available; residents become irritated and perplexed when HOAs fail to regularly update their lists; and residents don't always complain about the service providers they've selected from the list to the HOAs, but they might get resentful against each other without the HOAs knowing.

Finally, the listing of houses was examined. Today, brokers list on many different platforms, on an HOA portal and sometimes even a printed PDF flyer. Owners/Residents wanting to sell or rent their homes often post on local bulletin boards or WhatsApp groups. We observed that brokers then need to input the same data into multiple systems using a tedious process with high potential for human error. Before a showing, brokers detest phoning the HOA front desk to obtain the gate access code or the most recent parking regulations. They would never bring up this remark in an interview since they believe it is "just part of the job." Often times they guess, get it wrong and are shouted at by security staff. You notice properties listed as “For Sale” on the HOA's private property remain there for 3-6 weeks after closing. This is a nuisance for the broker who looks like they are not professional and for the residents who get asked constantly about the units that are not available.

Participatory:

VALORA says participation is the engagement and interaction of all user categories (residents, homeowners associations, estate agents, investors, visitors) in an active way all with the objective of mutual benefit. It is a two-way street where users give information and get value, and where users collaborate to improve community functioning, instead of being passive recipients of information.

The observational study is the first step in this process. Here, we audit existing systems, and shadow and observe the various user kinds. We begin by shadowing the email overflow, compliance triage and vendor management initially with the HOA administrators. We learn how residents today complain, find providers, and engage in interaction while shadowing residents. We can see listing procedures, data duplication and access problems by shadowing

estate agents. Lastly, we audit existing systems such as Property24 or Trafalga, Excel sheets, WhatsApp groups and even physical binders. The findings can thereafter be used for the design of a workshop in priority. During the session we can ask the residents to differentiate between features that can be added in the future and features that are essential to the MVP. In order to map features to actual user journeys, we can carry out story mapping with residents and the staff. For their present problems, the residents are able to validate or deny our observational findings. Residents and the team can help ensure the technical team is informed of resident priorities to help line up the vision. The third phase which is validation meetings are followed up. Low fidelity wireframes can be created based on the workshop priorities, then a prototype walkthrough can be done with the residents to apply their feedback. The workflows of the provider management and complaint moderation for the HOA administrator can be verified. Property intelligence routines and listing routines are validated in the agent feedback session. In addition, ROI and rental yield analytics requirements are also verified at the Investor feedback session. The continuous feedback loop carried out during development is the last stage. This may be sustained by holding sprint reviews every two weeks to talk about progress and get input that is consistent with Agile Scrum. Testing conducted in a real-world setting in one complex before the full launch is called Beta Testing, and this is done by a limited number of residents. Ultimately, the entire team is part of the feedback loop, a dedicated Slack or WhatsApp channel for continuous feedback.

Interviews:

In order to gather information from users of the current real estate and property management market to help better the user experience of VALORA, we conducted a structured interview. We Interviewed five people, where each individual person represented the main users of VALORA, such as the Resident, Investor, Estate Agent, Casual Visitor and HOA Admin. These are the set questions that we asked each user:

- **Question 1:** Out of these four platforms, Property24, Private Property, Trafalgar and EstateMate, which platforms have you used and what did you like the most and least about it?
- **Question 2:** When considering buying or investing in a property, would you want to be able to get a free AI valuation of the property as well as an estimate on your return on investment (ROI)?
- **Question 3:** How do you currently log or view complaints, request maintenance, look for service providers or receive communication within your community? Do you face any difficulties or frustration in doing so?
- **Question 4:** How would you feel about receiving communication, such as community announcements on one platform rather than multiple different ones?
- **Question 5:** Do you have any concerns about using a platform that stores your personal information, complaint history and property data in one place?

These were the responses that we received from each interviewee:

Interviewee 1: Resident

This Resident has used Private Property and has tried to use EstateMate, and they stated that Private Property was useful for browsing properties but it lacked information about the actual community and future valuations of the properties. They also found that EstateMate has good features however, it does not function properly. They stated that receiving an AI valuation and ROI estimate would help them to understand if the property that they possibly purchase has a growing value. Currently, the Resident uses WhatsApp to report any issues to the HOA and they find it frustrating as the HOA sometimes takes too long to respond to messages or even just doesn't respond at all, and for service providers they just ask their family, friends or neighbours for recommendations and contact numbers. They also stated that they would love to receive feedback from one platform rather than multiple. They are concerned that their personal and private information would be leaked but are fine if only HOA Admins can view that information.

Interviewee 2: Investor

We interviewed an Investor and they stated that they use Property24 and Private Property. They like that there are many listing and search filters, but dislike that there is nothing about community data. They also feel that an AI valuation and ROI estimate would save them much time, instead of them doing it themselves on excel. The investor believes that having a sufficient amount of information about the area and having the ROI estimate would really help them, as well as buyers in making wise purchase decisions. They currently don't log complaints, but they do look at community forums and ask residents about issues. They believe that having both property intelligence and community data in one place would make their investment decisions much easier. They aren't too concerned about their data but would want the platform to be transparent about how data is used.

Interviewee 3: Estate Agent

We interviewed an Estate Agent that lists properties on Property24 and Private Property, and found that they like how properties are able to be viewed easily, however they dislike that both Property24 and Private Property are expensive to use, and they don't show any community features that possible buyers ask about. They also thought that the AI valuation and ROI estimation would be useful to justify their listing prices and would attract more buyers if they knew the financial growth of their property. The Estate Agent does not really need to log any complaints. They feel that having all the property and community data in one place would make their job much easier, as they would be able to show possible buyers everything about a community without needing to switch between different platforms. They also stated that as long as the platform is transparent about their security, and the data is encrypted and protected they feel comfortable about trusting it for use.

Interviewee 4: Casual Visitor

We interviewed a Casual Visitor who just browses property listings and is interested in learning more about property values, but they are not ready to buy yet. They stated that they've used Property24 for browsing a few times and like how easy it is to search, however they dislike that there is too much information. They also said that they would find the AI valuation and ROI estimate useful as it would help them to understand if the properties are fairly priced before they actually consider buying a property in the future. They said that they didn't need to log complaints because they don't live in a gated community yet. They feel that a single platform for communication would be easier to use as they wouldn't need to create multiple accounts and passwords for different platforms. As a Casual Visitor they want to only be able to provide basic information such as an email address, but wouldn't want to provide any other information.

Interviewee 5: HOA Admin

We interviewed an HOA Admin who manages a gated community, and they stated that they've used Trafalgar for some financial management and EstateMate for resident communication. They found that Trafalgar was good for levies and financial reports but it was not intended for resident use, and also that EstateMate was good for communication and electronic billing, but didn't have any property investment features. They believe that an AI valuation and ROI estimate would be useful for the residents and investors, but not really for the HOA Admin. They currently receive complaints through WhatsApp, email and phone calls which can be missed easily when there is a high number of complaints coming in at once. The service providers are basically managed through a paper list of contacts as well as word of mouth. They agree that one platform for communication would be much simpler to use as it would reduce their workload significantly and make everything more organised. They feel that since they handle sensitive resident data, keeping that data safe is their key priority. They would need the platform to be POPI Act compliant, with encryption and clear data usage policies.

Analysis of Existing System

There are a number of property management and real estate platforms that are already available in South Africa. These systems have great value to buyers, sellers, estate agents, and property managers. Each platform, however, has a certain functionality and cannot deliver all of the functionality that VALORA would like to offer.

Property24 is among the most popular property websites in South Africa. It allows users to search for residential and commercial properties for sale or rent using filters such as price, location, and property type. If users want to know more than just what's in the property listing, they can also view suburb information, nearby schools, local property trends, and an estimate of their monthly bond repayments. These features help buyers make more informed decisions before purchasing a property, but the main point of Property24 is to have property

listings. It offers little investment knowledge and no tools such as AI property valuation, long term return forecasts, complaint tracking, and community management tools.

Another popular property listing site is **Private Property**, which helps buyers, sellers, landlords and estate agents connect with one another. This website is user-friendly and offers the ability to search for properties available throughout South Africa. It also includes market information, property alerts, and more to keep users updated on new properties. But like Property24, it is primarily used for property advertising. It does not offer residents any tools to report problems, to deal with complaints or access community services in private estates.

Trafalgar is more about the property management and not the sales of property. It offers services like body corporate management, financial management, and rental management for residential and commercial buildings. This can be beneficial to property owners and managing agents. But, it primarily targets property management professionals and organizations and not the general public or investors. It doesn't include a community platform for residents to report issues or to generate an AI-powered analysis of the property.

EstateMate is intended to help estate and community managers to manage residents and visitors, communicate with them, and to oversee the day-to-day activity on the estate. It improves communication in communities, but it doesn't provide in-depth property intelligence or investment tools. Other platforms would still be required to get property valuations, rental yield estimates or long-time investment information from users.

These systems all offer helpful features but only address a portion of the issue. These are the platforms that buyers typically need to search for homes to purchase, analyze the investment potential of the homes they are interested in, and connect with the community where they will reside. The goal of VALORA is to combine all of these services into one system, which will include community management, complaint tracking, property listings, AI-powered property analytics, and access to trusted service providers. This ensures that users do not need to use different platforms and have a more comprehensive solution.

Data Analysis (Data Integrity s Constraints)

Data integrity and constraints are used to ensure that the data stored in a database is accurate, consistent and reliable which are key principles to our proposed system. Unlike public listing platforms, VALORA handles sensitive resident information, HOA governance, complaint tracking and AI-driven financial estimates. The analysis phase helps us to see the ways in which existing systems maintain data integrity, define the constraints for VALORA’s PostgreSQL database, and to justify why VALORA’s approach is stricter and better suited for a private gated community.

Property24 and **Private Property** are two of South Africa's leading property listing platforms. Their data integrity strategies are built to make sure that listings are complete and that basic entry errors are not made. Entity Integrity makes sure that every listing has a unique listing_ID. Referential Integrity links a property to a valid agent_ID. Domain Integrity enforces data types such as price (decimal) and bedrooms (integer). It also mandates fields like location and price as NOT NULL. Property24 and Private Property, however, lack Row-Level Security as all listings are public, resident identity verification and complaint accountability features.

Trafalgar is a professional property management platform managing thousands of residential units in many buildings nationally. Unlike public listing portals Trafalgar provides full-service HOA and sectional title management, which includes online portals for levy accounts and maintenance requests. They have very strict financial constraints with annual budgets, levy schedules and audit compliance, and they have implemented an AI chatbot and Google Gemini AI to have instant access to financial reports and legal documents. Their systems, however, are made for managing agents and trustees, not for residents that want property investment intelligence or community complaint tracking.

EstateMate is a mobile community management platform designed for residential estates to make communication between homeowners, body corporates and managing agents easier. It links users to specific units which ensures entity integrity, referencing specific residents in maintenance tickets and facility bookings maintaining referential integrity, and restricting facility booking availability enforces domain integrity. EstateMate is POPI Act compliant and has role-based access for residents, trustees and property managers. It focuses on daily operations and communication, where it therefore lacks property valuation intelligence, AI-driven ROI projections and estate agent listing capabilities.

Data Integrity Constraints for VALORA:

VALORA's database supports property intelligence, community management and user management. These constraints ensure data accuracy, consistency and security for the platform:

Entity Integrity (Unique Identifiers):

- **Users Table:** user_id (UUID) is the Primary Key.
- **Properties Table:** Composite Primary Key: (complex_name + unit_no) ensures that each unit exists only once.
- **Complaints Table:** complaint_id (auto-generated) acts as the Primary Key.
- **Service Providers Table:** provider_id (UUID) serves as the Primary Key.

Referential Integrity (Foreign Keys):

- **User → Complaints:** complaints.user_id references users.user_id 'ON DELETE CASCADE
- **Property → Complaints:** complaints.unit_id references properties.unit_id 'ON DELETE CASCADE
- **Complaint → Work order:** maintenance_work_orders.complaint_id references complaints.complaint_id 'ON DELETE CASCADE'
- **Provider → Work order:** maintenance_work_orders.provider_id references service_providers.provider_id 'ON DELETE CASCADE'
- **User → Reviews:** reviews.user_id references users.user_id 'ON DELETE CASCADE'
- **Provider → Reviews:** reviews.provider_id references service_providers.provider_id 'ON DELETE CASCADE'
- **Property → Sales:** sales.unit_id references properties.unit_id 'ON DELETE CASCADE'
- **Property → Listings:** listings.unit_id references properties.unit_id 'ON DELETE CASCADE'
- **User (Agent) → Listings:** listings.unit_id references users.user_id 'ON DELETE CASCADE'
- **User (Admin) → Announcements:** announcements.created_by references users.user_id 'ON DELETE CASCADE'
- **Property → AI analytics:** ai_analytics.property_id references properties.unit_id 'ON DELETE CASCADE'
- **User → Sessions:** sessions.user_id references users.user_id 'ON DELETE CASCADE'

Domain Integrity (Check Constraints):

Table	Column	Constraint
properties	size_m2, bedrooms, bathrooms	CHECK (> 0) or CHECK (>= 0)
listings	price	CHECK (price numeric > 0)
ai_analytics	rental_yield_pct	CHECK (BETWEEN 0.00 AND 100.00)
complaints	status	CHECK: ('submitted', 'acknowledged', 'in_progress', 'resolved', 'cancelled')
complaints	category	CHECK: ('sanitation', 'utilities', 'security', 'infrastructure', 'noise', 'other'
service_providers	avg_rating	CHECK (BETWEEN 1.0 AND 5.0)
users	role	ENUM: ('casual visitor', 'resident', 'investor', 'agent', 'admin')

Mandatory Fields (NOT NULL):

- **User Registration:** user_id, email, Password_Hash, role
- **Properties :** unit_id, unit_no, street_address
- **Sales:** sale_id, unit_id, price, sale_date
- **Complaint Logging:** complaint_id, user_id, unit_id, category, description
- **Service Provider:** provider_id, full_name, trade, phone
- **Maintenance Work Orders:** work_order_id, complaint_id
- **Reviews:** review_id, user_id, provider_id, rating
- **Listings:** listing_id, unit_id, agent_id, price
- **Announcements:** announcement_id, title, message, created_by
- **AI Analytics:** analytics_id
- **Sessions:** session_id, user_id, refresh_token, expires_at
- **Login Attempts:** attempt_id

VALORA has some complex constraints that are unique to the platform. For the **Geographic Boundary,** Property24 is for national listings, whereas VALORA doesn't implement this as it restricts it to a single complex. **Row-Level Security (RLS)** is where residents view only their own unit, and investors see aggregated complaint trends but no resident names. This ensures that we comply with the POPI Act and protect the privacy of residents, which bypasses EstateMate's role-based access. **Financial Visibility** refers to allowing only the Investors and Estate Agents the ability to view the ROI and rental yield. This helps to protect any sensitive financial information. **Automated Complaint Routing** is where the database trigger assigns complaints to the HOA Admin based on the category. This automates the workflow and is similar to EstateMate's ticketing except that VALORA uses AI-driven classification. **AI Training Data Integrity** utilizes CHECK which ensures that the sale_date is within the last 10 years. This prevents the XGBoost model from training on outdated data. **Unique Review Constraint** is the composite UNIQUE (unit_id, provider_id) that ensures one review per provider. This helps to prevent review spam and ensures authentic feedback.

VALORA's approach to data integrity combines the professional governance focus of Trafalgar, the community engagement applications of EstateMate and property listing capabilities of Property24/Private Property, but with a stricter set of constraints. By implementing composite primary keys, referential integrity with restrict actions, row-level security, automated triggers for complaint routing and AI training data validation, VALORA brings data accuracy, resident privacy and operational efficiency to AI-driven property valuations and community management.

Weakness of Current System

We analyzed four property management systems to understand where their strengths and weaknesses lie. These systems were chosen as they are the two primary segments of real estate and community/property management that are included in the VALORA Private Property and Community Management System. Even if each of them excels in their particular field, no single system offers end-to-end capabilities that include community management, property management, property listings, maintenance tracking, and intelligent property analysis all in one platform. The analysis identified the strengths and weaknesses below.

Property24:

Property24 is South Africa's biggest online property marketplace. It allows the user to search for both residential and commercial property, filter results to narrow down search, contact estate agents and advertise property for rent or sale. It allows for a large property database for homes and businesses; advanced search options by location, price, property type and number of bedrooms; professional property pictures and descriptions; easy communication between the estate agent and the buyer; and property notifications for registered users.

Despite its popularity, there are some drawbacks of Property24 such as the platform tends to emphasize primarily property ads and listings instead of continual management of properties. There are no maintenance request or fault reporting features that are built in. Residents cannot submit complaints or communicate with the property management within the platform. It does not offer any AI property valuation, rental yield analysis or investment prediction. There are no features for community management like announcements, resident communication and service provider management.

Private Property:

Private Property is a site that provides functionality similar to Property24 in enabling people to search, purchase, sell or rent homes and property all over South Africa. It delivers a user-friendly interface and information on the properties. It allows for in-depth property search options, a wide selection of residential and commercial properties, firsthand contact with estate agents, and property alerts and saved searches.

However, there are some drawbacks to the platform as it emphasizes the marketing of property and not administration; it does not include maintenance management; there is no complaint system in place; managing a HOA is not supported; and AI property valuation, rental yield analysis, growth score analysis, and other investment decision tools are not available.

Trafalgar:

Trafalgar offers a wide range of property management services such as sectional title management, HOA management, rental management and financial administration services. It also enables residents and owners to view the property and make requests for maintenance. It provides their users with proper management of the property and estate, web-based maintenance claim system, financial management and reporting, and Owners and trustees' documentation. Finally, it facilitates sectional title and HOA administration.

Although Trafalgar offers a number of good property management features, there are some it will not work with. The site is not a full real estate website to purchase and sell properties. It lacks property valuation, and you cannot receive an investment analysis via AI. There are limited community involvement elements in this system, while managed properties are used by some services and may not be available for all services. It does not have a single platform to handle all stakeholder services.

EstateMate:

EstateMate is a community management and communication platform used by gated estates and complexes in South Africa. It brings about easier communication between residents and property managers, commercial tenants and facility managers.

Even though EstateMate has many community management features, it’s weakness is what make it unpleasant for user interaction. Their interface design is misleading as it states that it is a mobile application, yet it cannot be properly viewed online by a mobile device. There are frequent reports made by users about bugs in the mobile application which causes the system to crash. It lacks the property functionality that many users want access to. The platform also charges a fee per property, and in the case of larger complexes or estates it increases the levy amount for the residents.

Comparison with the Proposed VALORA System:

The analysis shows that the current systems are developed to address certain problems, but not a full property ecosystem. Property24 and Private Property are good real estate websites, and Trafalgar and EstateMate are good community property management websites. However, none of these systems have a single platform that encompasses all the necessary services.

The VALORA Private Property and Community Management System aims to overcome these drawbacks by having all the listed properties, property valuation using AI, analysis of rental yield and return on investment (ROI), maintenance request tracking, complaint management, community announcements, service provider management, homeowner association support, and secure communication all in one place. These capabilities are

combined to make VALORA a single system that allows users to manage all parts of property ownership and community living without having to switch between several systems. VALORA would therefore offer a more efficient, user-friendly and comprehensive solution for the residents, investors, estate agents, casual visitors and HOA admins.

The comparative analysis confirms that all of the existing systems offer valuable services, but they lack features that make them the "whole solution" for property and community management. The proposed VALORA system utilizes the benefits of these systems while mitigating their limitations by incorporating intelligent technologies, centralized data management, and improved communication capabilities. This holistic approach makes VALORA a more effective solution for today's property management and community administration.

Analysis of the Proposed System (Functional Requirements)

Functional Requirements define the specific features of the system. VALORA's main functional requirements include the Property Intelligence Module and the Community Management Module.

With the **Property Intelligence Module**, the system will be secured with unique credentials, and it will dynamically restrict features available to each user based on the five different roles, such as Casual Visitor, Resident, Investor, Estate Agent and HOA Admin. It will include automated ML valuation, where the system takes in user inputs of property parameters (unit square footage, number of bedrooms/bathrooms, localized suburb name) and returns a real-time valuation estimate based on the built-in Scikit-learn/XGBoost model. For registered Investors, premium investor analytics provide the mathematical projections that are calculated and outputted to include a 10-year ROI curve and estimated rental yield percentages. The estate agent listing interface will include a Create, Read, Update and Delete (CRUD) interface for verified estate agents to add, view, edit and remove a property listed within the community geographical limits.

For the **Community Management Module**, resident complaint logging & tracking will allow a registered resident to log an infrastructure or community issue and see the status (Submitted, Acknowledged, In Progress, Resolved, Cancelled) updated in real-time. AI Natural Language Classification would analyze the content of the resident's reported complaint and apply a text classification category tag (Utilities, Security, Noise, etc.) before the complaint is manually reviewed by an administrator. The system will provide a secure dashboard for HOA admins to moderate incoming complaints, update tracking statuses and send community-wide alerts or notices. A verified service provider directory will enable the HOA admin to add, subtract and verify entries to the list of local service providers that can be displayed to the residents of the community. The system will also enable residents to give the service providers a rating (1-star to 5-star and a written review) after they have been hired.

Non-Functional Requirements

The Non-Functional Requirements specify how the system should behave or operate.

For the **Performance**, FastAPI Core Latency will allow the Python backend to respond to complicated analytical queries and deliver ML evaluation scores within 2 seconds when traffic peaks. The user interface will be developed in React and designed to optimize the rendering of the resources so that a complete page can be displayed within 3 seconds on average on a standard mobile and desktop connection in South Africa.

For **Security and Availability**, the platform will implement role-based access tokens with JSON Web Tokens, which is used on all API requests, and the sensitive resident profiles and data fields will be encrypted at rest in the database. The platform architecture will ensure a 99.5% deployment uptime, with the ability to scale up automatically when there is an unexpected traffic surge, via AWS auto-scaling configurations.

The **Usability** of the website would be that it is fully responsive (works on desktop and mobile devices). The system would contain consistent navigation and be accessible to users with basic computer literacy. All complaint forms would have labels and a message when there is an error.

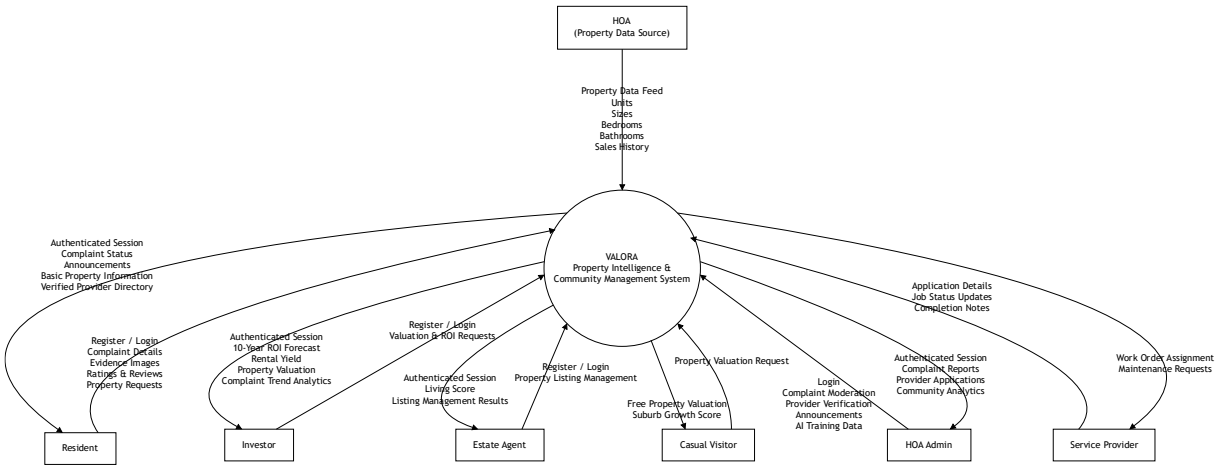
With the **User Interface**, the Visitor view will include searching and displaying visitor valuations for public property. The Resident dashboard will provide access to the complaint form, complaint list, service directory, and announcements. The Investor view would offer comprehensive investment reports and property short listings. The Agent dashboard displays listing management and inquiry log. The Complaint moderation would be where the provider management and analytics features are available in the HOA Admin Panel.

For the **Data Source**, the HOA would provide all property data such as units, sizes and sales history in a spreadsheet or API. There does not need to be any external property portals scraped.

The **Constraints** would be that the platform is to be delivered by November 2026. It will only be available for use at a single housing complex, and would not be available for use at multiple complexes in MVP. The development team is made up of nine specializing members. There is no external money, and hosting costs will be low cost or free (Hetzner low-cost VPS, AWS free tier).

Data Modelling for Proposed System

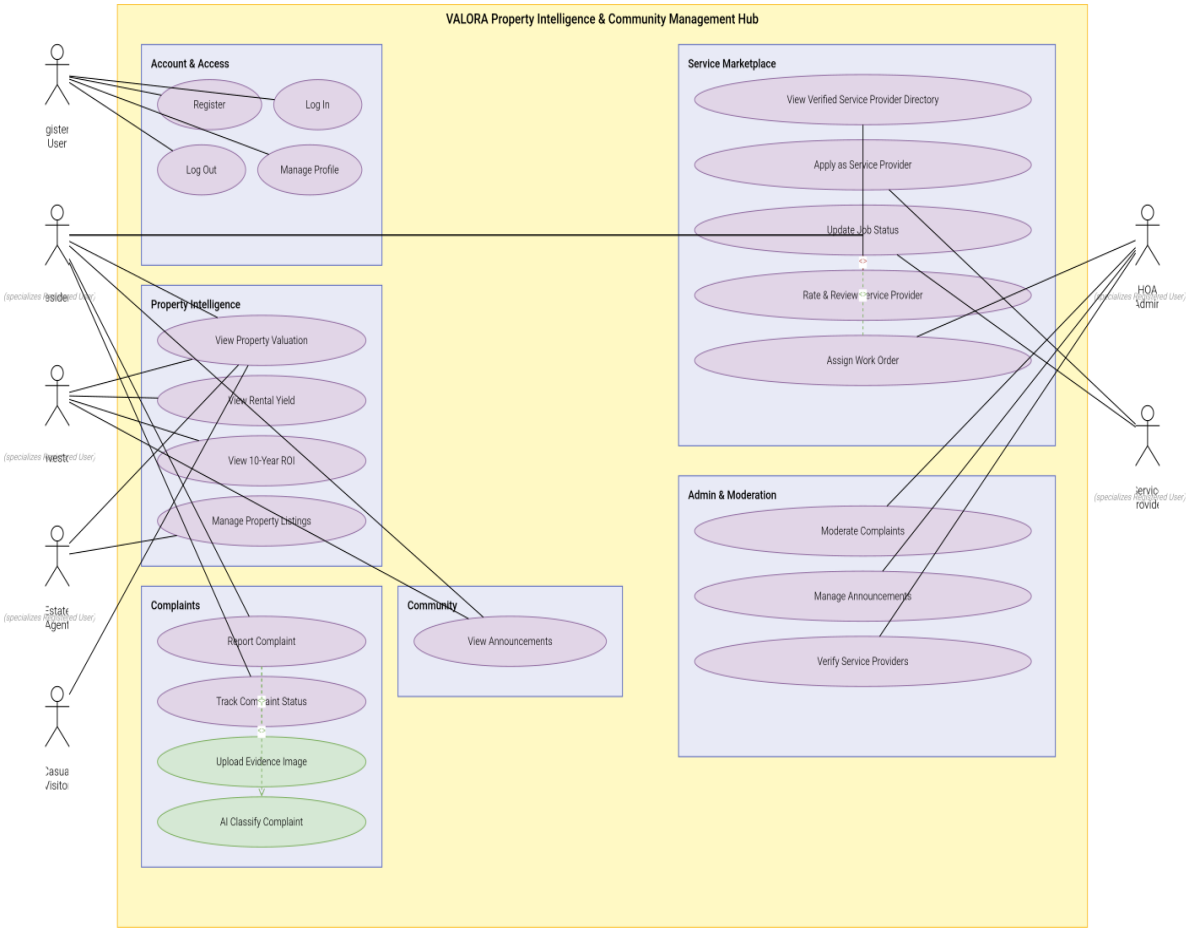
1. Context Level 0 Diagram:



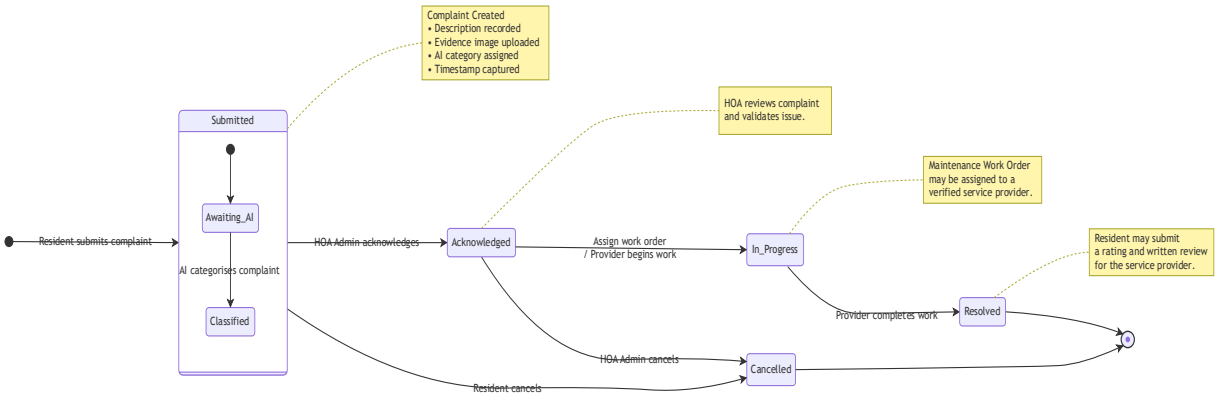
2. Entity Relationship Diagram:



3. Use Case Diagram:



4. Complaint State Diagram:



5. Data Flow Diagram:

